

Realized Returns to Green Investing: A Global Replication

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Abstract

We examine whether realized stock returns provide reliable evidence on the expected returns to green investing. The literature reports conflicting findings, with both significant green and brown outperformance. We conduct a comprehensive scientific replication that spans a broad set of greenness measures, portfolio constructions, risk models, and countries, covering the ex ante plausible empirical design space. Across these specifications, realized green-minus-brown returns are statistically insignificant once we take multiple testing into account. The distribution of published t -statistics differs significantly from the replicated distribution, consistent with publication bias. Finally, we show that realized returns are too noisy and samples too short to identify the greenium.

Keywords: Greenium, cost of capital, ESG, replication, climate, sustainable finance
JEL Codes: G11, G12, H23, Q5

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Returns to green investing are central to investors, firms’ incentives to make a green transition, and the academic literature on sustainable finance. However, the academic evidence is strikingly inconsistent. As seen in the solid blue bars of Figure 1, some studies report that green stocks earn significantly higher returns than brown stocks,¹ others find that brown stocks earn higher returns,² and still others find no significant difference.³ These papers often appear in leading journals, are highly cited, and their methods and data are widely used in the financial industry. Yet these papers reach opposite conclusions about the sign, magnitude, and even the existence of a greenium. This lack of agreement is puzzling, particularly given the shared goal: measuring the return difference between green and brown stocks.

In this paper, we argue that the conflicting evidence is not mysterious. Instead, it reflects the interaction of three basic challenges: (i) There is no single, agreed-upon measure of corporate greenness, as researchers use emissions intensity, emissions levels, ESG scores, sub-scores focused on the environmental part, toxic releases, or composite indicators; (ii) different papers use different methodologies, that is, samples, industry adjustments, and risk controls; and (iii) the literature evaluates many closely related specifications without systematically accounting for multiple testing, creating scope for chance findings to appear significant in classic single-hypothesis testing.

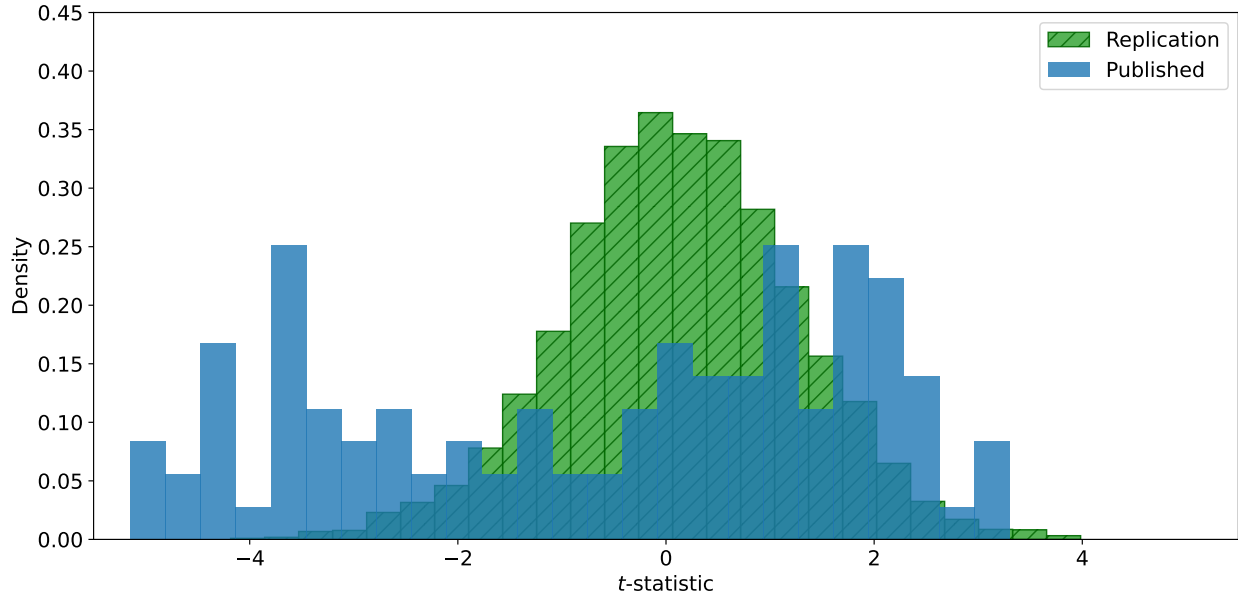
We have three main findings. First, we show that the realized returns of green-minus-brown portfolios are globally insignificant when we address the challenges listed above, namely we (i) use all greenness measures, (ii) combine them with all methodologies, and (iii) adjust for multiple testing. Second, we provide evidence that the literature may suffer from publication bias. Third, we show that the source of the problem is that the available realized return data are too short and too noisy to make statistical inference about expected

¹Garvey, Iyer, and Nash (2018); In, Park, and Monk (2019); Cheema-Fox, LaPerla, Serafeim, Turkington, and Wang (2021a,b); Giese, Nagy, and Rauis (2021); Huij, Dries, Stork, and Zwinkels (2021); Ardia, Bluteau, Boudt, and Inghelbrecht (2022); Bauer, Huber, Rudebusch, and Wilms (2022); Pástor, Stambaugh, and Taylor (2022); Zhang (2025); Berg, Lo, Rigobon, Singh, and Zhang (2023); Karolyi, Wu, and Xiong (2023).

²Alessi, Ossola, and Panzica (2020); Bolton and Kacperczyk (2021, 2023); Hsu, Li, and Tsou (2023).

³Görgen, Jacob, Nerlinger, Riordan, Rohleder, and Wilkens (2020); Pedersen, Fitzgibbons, and Pomorski (2021); Aswani, Raghunandan, and Rajgopal (2024); Alves, Krüger, and van Dijk (2025); Lindsey, Pruitt, and Schiller (2023).

Figure 1: Distribution of t -statistics of green-minus-brown portfolios globally



The blue solid bars show the distribution of published t -statistics from the literature on the alphas of global green-minus-brown portfolios. Similarly, the green striped bars show the distribution of our replicated t -statistics across all combinations of greenness measures, portfolio constructions, countries, and risk controls. A positive t -statistic indicates that green stocks outperform, where numbers above 1.96 are typically considered statistically significant (i.e., without multiple testing correction), and vice versa for negative numbers.

returns.

To illustrate the state of the literature, Figure 1 shows the distribution of published t -statistics. As seen in the figure, the distribution is bimodal, meaning that the literature has a disproportionately high frequency of significantly positive and negative t -statistics relative to the lower frequency of insignificant test statistics in the middle of the distribution. This could indicate a publication bias: researchers have looked across different specifications and the significant findings are more likely to be published.

To address publication bias, we examine an ex ante (pre-publication bias) distribution of t -statistics that arises when we systematically combine and intersect different methods from the literature. Moreover, we account for multiple testing. Constructing the full distribution of ex ante plausible t -statistics is central to understanding what can be learned from realized returns, since the literature might suffer from publication bias even when individual results

are reproducible.

Specifically, to assess what can be learned from realized returns, we conduct a comprehensive scientific replication of the green investing literature as follows. We construct green-minus-brown portfolios using a broad set of 23 greenness measures that we identify from the literature. For each measure, we consider alternative portfolio constructions, either industry-neutral (used by some papers) or industry-agnostic (used by other papers) green-minus-brown portfolios. Further, for each portfolio, we evaluate the performance using any of the standard risk adjustments used in the literature (average excess returns, CAPM alphas, Fama-French three-factor alphas, etc.). Lastly, we evaluate the performance in the United States as well as in 48 additional countries.

Our first and most central finding is simple: Across measures, models, and markets, realized return differences between green and brown stocks are not statistically different from zero once we control for multiple testing. This complete lack of significance is a striking finding for such a prominent literature, even in the context of replication studies more broadly.

The distribution of replicated test statistics is shown as the green striped bars in Figure 1. As seen in the figure, the distribution is centered near zero and has the bell shape that one would expect from noise. Some of the replicated t -statistics are significant at conventional levels, that is, larger than 1.96 or lower than -1.96 , but this is expected to happen in 5% of the tests by chance. In fact, about 8.6% of the replicated portfolios are larger than 1.96 in absolute magnitude, and this modest number of test statistics above 1.96 in magnitude explains why all these portfolio returns are insignificant when using the Benjamini-Hochberg multiple-testing correction.

Our second finding is to compare the published and replicated distributions in Figure 1. The literature’s distribution appears to be over-sampled in the left and right tails. In fact, we can statistically reject the null hypothesis that these distributions are identical with a Kolmogorov-Smirnov test (p -value < 0.001). This significant difference in distributions is surprising given that both supposedly capture the universe of greenium t -statistics and is

consistent with publication bias. Further, the literature’s distribution contains a particularly extreme left tail that does not overlap with our replication distribution, consistent with part of the literature using ESG measures with look-ahead bias as discussed by [Zhang \(2025\)](#).

Our final and third result seeks to shed further light on the source of these widespread robustness issues. We estimate that a GMB factor has an annual Sharpe ratio of at most -0.15 , computed as the ratio of the modest greenium estimated using forward-looking returns from [Eskildsen, Ibert, Jensen, and Pedersen \(2026\)](#), divided by the nontrivial realized GMB volatility. This low Sharpe ratio implies that one needs more than a century of realized returns to identify the greenium. Hence, a lack of robustness is not surprising given that the literature is generally based on less than 20 years of data, and often not much more than a decade.

Our paper is related to several literatures. First, we show that the literature on the realized returns to green investing rests on a shaky footing. While it might seem obvious in hindsight that 10-20 years of evidence is too short, these papers are nevertheless published in top journals as they deal with important issues and report significant results. In other words, their sample lengths are long enough to yield significant results with single-hypothesis testing. In fact, these papers are among the most cited recent papers in finance.⁴ We contribute by providing evidence on (i) the overall distribution of published findings, (ii) the full distribution of plausible performance metrics, controlling for multiple testing; and (iii) the needed sample length.

Second, we contribute to the rising literature on replication more broadly ([Hamermesh, 2007](#); [Welch, 2019](#)). Several fields have been accused of replication crises, including medicine ([Ioannidis, 2005](#)), psychology ([Nosek, Spies, and Motyl, 2012](#)), management ([Bettis, 2012](#)), and experimental economics ([Maniadis, Tufano, and List, 2017](#)). In finance, the replication rate has been found to be low for equity factors ([Hou, Xue, and Zhang, 2020](#); [Harvey, Liu, and Zhu, 2016](#)) and corporate bond factors ([Dick-Nielsen, Feldhütter, Pedersen, and](#)

⁴E.g., two of the top-three most cited recently in the *Journal of Finance* are about the greenium ([Bolton and Kacperczyk \(2023\)](#); [Hsu et al. \(2023\)](#)) and so are four of the the top five from the *Journal of Financial Economics* ([Bolton and Kacperczyk \(2021\)](#); [Flammer \(2021\)](#); [Pástor, Stambaugh, and Taylor \(2021\)](#); [Pedersen et al. \(2021\)](#)); websites accessed 2/22/2026.

Stolborg, 2023), although the former result has been overturned (Chen and Zimmermann, 2020; Jensen, Kelly, and Pedersen, 2023). Conceptually, our approach is similar to research in asset pricing that algorithmically generates a universe of factors to control for publication bias (see, e.g., Yan and Zheng, 2017; Chordia, Goyal, and Saretto, 2020). Our result that all findings on GMB returns are insignificant with multiple-testing adjustment is unusual since the debate in other areas is often about whether or not the “majority” of papers remain significant when replicated.

Third, we contribute to sustainable finance more broadly. Our results do not imply that environmental concerns are irrelevant for asset prices or corporate behavior. Rather, they highlight the limits of using realized equity returns to measure small differences in expected returns over short horizons. Understanding how climate risks and preferences affect asset prices likely requires moving beyond realized returns toward theory, forward-looking measures, price reactions to identifiable shocks, and evidence on investor behavior and capital allocation. For example, if some investors prefer green stocks, theory suggests that these become expensive and therefore have low forward-looking returns (Pástor et al., 2021; Pedersen et al., 2021). Moreover, firms’ forward-looking perceived costs of capital also reflect that green stocks’ expected returns are modestly below those of brown ones (Gormsen, Huber, and Oh, 2023). Qualitatively similar conclusions can be reached using forward-looking expected returns based on prices and expected earnings (Chava, 2014; Pástor et al., 2022; Eskildsen et al., 2026), survey evidence (Giglio, Maggiori, Stroebe, Tan, Utkus, and Xu, 2025; Edmans, Gosling, and Jenter, 2024), and experiments (Bonnenfon, Landier, Sastry, and Thesmar, 2025; Heeb, Kölbel, Paetzold, and Zeisberger, 2023).

1 Greenness Measures and Data

We consider 23 greenness measures from the literature. Table 1 shows the greenness measures and the corresponding references. The green scores are based on data from Trucost, MSCI, Sustainalytics, and the Environmental Protection Agency (EPA) as shown in the

table.⁵ The EPA data are only available in the US, so our global ex-US sample contains 19 individual greenness measures.

We sign each greenness measure such that greener stocks have higher values. The 23 greenness measures include MSCI’s and Sustainalytics’ ESG scores as green investors may use these as catch-all sustainability metrics. For instance, 88% of the respondents in BlackRock’s 2020 Global Sustainable Investing Survey rank Environment as their top priority among the ESG factors.

Most of the greenness measures have been used to study the returns to green investing, but, for completeness, we also include a few that have been used in other types of studies. The idea to construct a comprehensive set of greenness measures that researchers could have plausibly considered ex ante to mitigate the effects of publication bias (i.e., the tendency of statistically significant results to be more likely to be published than insignificant results). For each measure, we seek to only use the data when they are available to investors and lag the data appropriately. Internet Appendix A.1 contains additional details.

We obtain realized stock returns and stock characteristics from the data set in [Jensen et al. \(2023\)](#) available through [WRDS](#). Monthly realized returns are from CRSP for U.S. stocks and from Compustat for non-U.S. stocks. We collect accounting data from Compustat, using quarterly information when it is available and annual information otherwise. Following [Jensen et al. \(2023\)](#), we restrict our sample to common stocks listed on the NYSE, NASDAQ, or AMEX in the United States and to common stocks listed on major exchanges outside the United States. For U.S. firms, we retain all available classes of common shares, whereas for non-U.S. firms we keep only the primary share identified by Compustat. US equity factor returns are from Kenneth French’s data library and from [global-q.org](#). Global ex-US equity factor returns are from [Jensen et al. \(2023\)](#).⁶

⁵Sustainalytics began transitioning from ESG scores to ESG risk ratings in 2018. We rely on the legacy ESG scores from 2009 until they are phased out at the end of 2019. To extend the series through December 2022, we reconstruct the scores using indicator-level data and weights from the updated framework, which closely mirrors the legacy methodology. In the period when both the original and reconstructed scores are available, the average correlation between them exceeds 80%.

⁶The [Jensen et al. \(2023\)](#) factors are available at [jkpfactors.com](#). We use the value-weighted market return and the capped value-weighted return of all non-market factors. The non-market factors are based on the following characteristics (with the factor name in parentheses): market equity (size), book-to-market

Table 1: Greenness measures

Name	Source	References
S1INT (Sales)	Trucost	Bolton and Kacperczyk (2021, 2023), Busch, Bassen, Lewandowski, and Sump (2022) ^c , Aswani et al. (2024), Atilgan, Demirtas, Edmans, and Gunaydin (2023), Zhang (2025)
S1+2INT (Sales)	Trucost	Griffin, Lont, and Sun (2017), Garvey et al. (2018) ^a , G6rgen et al. (2020) ^a , Cheema-Fox et al. (2021a), Cheema-Fox et al. (2021b), Giese et al. (2021), Huij et al. (2021), Pedersen et al. (2021), Bauer et al. (2022) ^a , Shakdwipee, Giese, and Nagy (2023)
S1+2+3INT (Sales)	Trucost	In et al. (2019), Cheema-Fox et al. (2021b), Ardia et al. (2022) ^{a,c} , Busch et al. (2022) ^c
S1INT (Assets)	Trucost	Shakdwipee et al. (2023) ^b
S1+2INT (Assets)	Trucost	Shakdwipee et al. (2023) ^b
S1+2+3INT (Assets)	Trucost	S&P Dow Jones Indices (2020) ^b
Weighted ESG score	MSCI	Ang, van Beek, Li, Tamoni, and Zhang (2023), Lindsey et al. (2023)
Environment score	MSCI	Engle, Giglio, Kelly, Lee, and Stroebel (2020), G6rgen et al. (2020), Berg et al. (2023), Lindsey et al. (2023)
Total ESG score	Sustainalytics	Alves et al. (2025), Lindsey et al. (2023)
Environmental score	Sustainalytics	Engle et al. (2020), G6rgen et al. (2020), Seltzer, Starks, and Zhu (2022), Alves et al. (2025), Lindsey et al. (2023)
LOG(S1TOT)	Trucost	Bolton and Kacperczyk (2021, 2023), Aswani et al. (2024), Atilgan et al. (2023), Zhang (2025)
LOG(S1+2TOT)	Trucost	Huij et al. (2021), Bauer et al. (2022) ^a
LOG(S1+2+3TOT)	Trucost	Matsumura, Prakash, and Vera-Mu6oz (2014) ^{a,b} , Delmas, Nairn-Birch, and Lim (2015) ^c , Busch et al. (2022) ^c
Ind.-adj. ESG score	MSCI	G6rgen et al. (2020), Pedersen et al. (2021), Alves et al. (2025), Ang et al. (2023), Berg, K6lbel, Pavlova, and Rigobon (2023), Berg et al. (2023), Lindsey et al. (2023)
Greenness (PST)	MSCI	P6stor et al. (2022), Karolyi et al. (2023)
E climate score	MSCI	Cheema-Fox et al. (2021b), Kacperczyk and Peydr6 (2022) ^c
E nat. res. score	MSCI	Kacperczyk and Peydr6 (2022) ^c
E waste score	MSCI	Kacperczyk and Peydr6 (2022) ^c
E env. opps. score	MSCI	Cheema-Fox et al. (2021b), Kacperczyk and Peydr6 (2022) ^c
TPWINT (Sales)	EPA TRI	Hsu et al. (2023)
TPWINT (Assets)	EPA TRI	Hsu et al. (2023)
TRINT (Sales)	EPA TRI	Akey and Appel (2021) ^d
TRINT (Assets)	EPA TRI	Akey and Appel (2021) ^d

The table shows data sources for 23 individual greenness measures. The table also shows the academic papers which use a particular greenness measure when studying realized financial performance, in particular realized stock returns. S1TOT, S1+2TOT, and S1+2+3TOT refer to the absolute amount of carbon emissions using scope 1, the sum of scope 1 and 2, and the sum of scope 1, 2, and 3 carbon emissions, respectively. S1INT, S1+2INT, and S1+2+3INT refer to the respective carbon intensities, i.e., total emissions scaled by sales or assets. Scope 3 refers only to upstream emissions as downstream emissions are only available from 2017. Greenness (PST) refers to the measure of P6stor et al. (2022). Ind.-adj. ESG score refers to MSCI’s industry-adjusted ESG score. E nat. res. score and E env. opps. score refer to MSCI’s natural resource and environmental opportunities scores. TRINT and TPWINT refer to toxic release intensity and toxic production waste intensity from the Environmental Protection Agency. The superscript ^a indicates a paper using carbon emissions, but from another data source than Trucost. The superscript ^b indicates references showing that practitioners and regulators also scale emissions by assets, typically EVIC (enterprise value including cash), which we proxy for by book assets to avoid introducing biases by having market values on the right-hand side. The superscript ^c indicates a paper using a dependent variable other than realized stock returns. The superscript ^d indicates a paper using toxic releases (which are arguably more relevant for pollution than the toxic production waste used in Hsu et al. (2023)) and which we then scale as in Hsu et al. (2023).

We also consider the distribution of published t -statistics from the literature. We detail the collection of these t -statistics in the Internet Appendix. To be consistent with our scientific replication, we sign t -statistics such that a positive t -statistic indicates green out-equity (value), operating profit-to-book equity (profitability), asset growth (investment), and 12-1 month past returns (momentum).

performance over brown, whereas a negative t -statistic indicates brown outperformance over green.

2 Green-Minus-Brown Portfolios

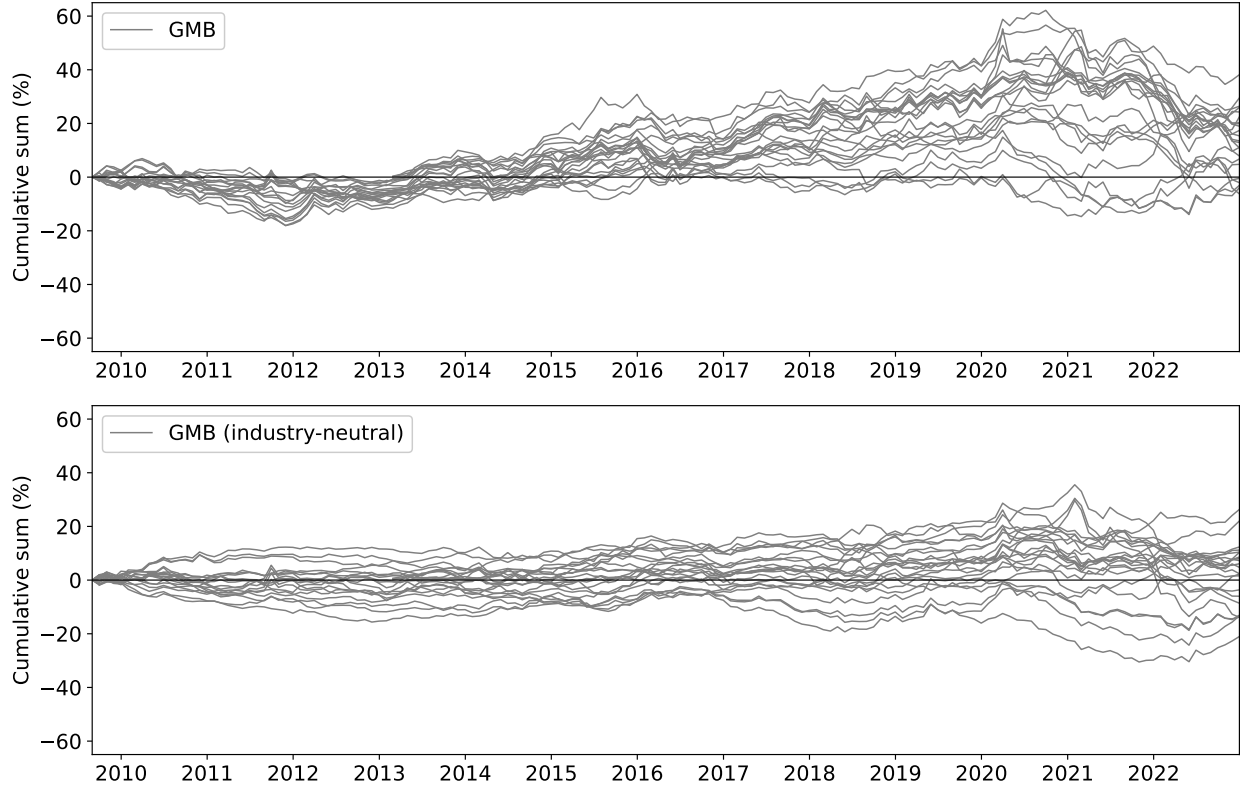
For each of the 23 greenness measures, we construct an industry-neutral and an industry-agnostic green-minus-brown portfolio. This process yields $2 \times 23 = 46$ GMB factors. We use this method to encompass both the papers in the literature that focus on industry-agnostic portfolios (e.g., [Pástor et al., 2022](#); [Zhang, 2025](#)) and those that focus on within-industry variation (e.g., [Bolton and Kacperczyk, 2021, 2023](#); [Hsu et al., 2023](#)). Both approaches are reasonable ex ante—the decision of whether to adjust for industry reflects the usual trade-off between mitigating omitted variable bias at the cost of eliminating potentially interesting variation.

Starting with the US, to construct each industry-agnostic GMB factor, we sort US stocks into terciles each month according to each greenness measure. We then compute next month’s portfolio return for each tercile by value-weighting stocks with a cap on market capitalization at the NYSE 80th percentile, as in [Jensen et al. \(2023\)](#).

Likewise, to construct industry-neutral GMB factors, we first sort stocks into terciles within each industry, then combine these terciles across industries, and then compute value-weighted capped returns. Finally, we compute a GMB portfolio return for each greenness measure as the return difference between the top tercile (the green portfolio return) and the bottom tercile (the brown portfolio return).

Figure 2 shows the cumulative returns of the 46 GMB portfolios that we construct based on the 23 greenness measures in the US sample. As seen in the figure, the returns are noisy over a short sample period and vary across the greenness measures.

Figure 2: Replicated Green-Minus-Brown Portfolios in the US



The figure plots the cumulative sum of realized returns of 46 green-minus-brown (GMB) US equity factors constructed using 23 different greenness measures with either an industry-agnostic (top panel) or industry-neutral (bottom panel) approach. The time period is from September 2009 to December 2022.

3 Scientific Replication of the Literature: US Evidence

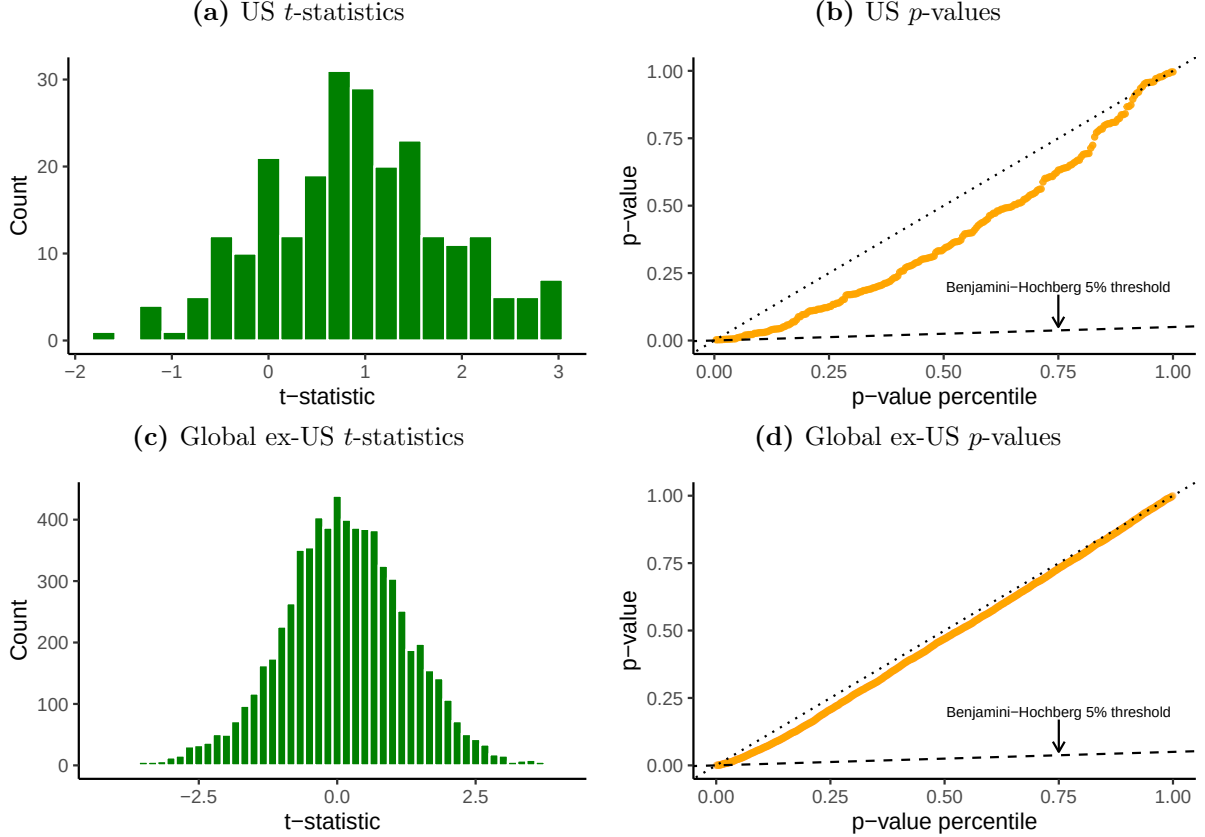
For each of the 46 factors, we compute the alpha in five different ways to account for risk exposures measured in different standard ways: 1) no risk adjustment (excess returns), 2) the CAPM, 3) the Fama-French three-factor model, 4) the Fama-French five-factor model augmented with momentum, and 5) the $q5$ -factor model (Hou, Xue, and Zhang, 2015, 2021).

Figure 3(a) plots a histogram of all t -statistics, corresponding to the $23 \times 2 \times 5$ alphas of the US GMB portfolios.⁷ The figure shows that most of the t -statistics are less than 1.96 in absolute value, meaning that the corresponding alphas are insignificant at the conventional 5% significance level. However, there are a number of larger t -statistics, which helps explain

⁷All the individual t -statistics are reported in the Internet Appendix, Tables A2 and A3.

why the literature sometimes finds a significant greenium. We show in this subsection that none of these larger t -statistics survive a multiple testing adjustment.

Figure 3: Distributions of t -statistics and p -values



Panels (a) and (c) show the distributions of t -statistics of all GMB alphas. In Panels (b) and (d), the solid line show the corresponding p -values ranked from low to high, plotted against their percentile rank on the x -axis. The dotted line is the 45-degree line and the dashed line shows the multiple testing threshold from [Benjamini and Hochberg \(1995\)](#) that controls the false discovery rate at a 5% level. Since the p -values in the solid line is above the dashed line, all of the estimates are statistically insignificant after adjusting for multiple testing. To compute these results, GMB factors are constructed using 23 (US) and 19 (Global ex-US) individual greenness measures, separately for the US and 48 other countries, with and without industry adjustment. We estimate alphas with respect to 1) no risk adjustment (excess returns), 2) the CAPM, 3) the Fama-French three-factor model, 4) the Fama-French five-factor model augmented by momentum, and, in case of a US GMB factor, 5) the $q5$ -factor model. For instance, Panel (a) contains $23 \times 2 \times 5 = 230$ t -statistics. Standard errors are [Newey and West \(1987\)](#) adjusted with three lags.

We compute the multiple-testing adjustment of [Benjamini and Hochberg \(1995\)](#), which is one of the most commonly used methods. The Benjamini-Hochberg (BH) method considers the p -values from all tests and classifies them as significant or insignificant in such way that

the false discovery rate (FDR) is controlled at 5%.

The BH method first ranks the p -values from low to high as seen in the solid line in Figure 3(b). This cumulative distribution of p -values is then compared to the threshold $m/M \times 0.05$ (dashed line), where m/M is the percentile rank of a given p -value (i.e., the rank, m , divided by the total number of tests, M). The BH method then finds the highest value of m for which the actual p -value is below this threshold and rejects the null hypothesis for this test and all tests that have a lower rank. For a textbook treatment of these issues, see [Efron \(2012\)](#).

As seen in Figure 3(b), the ranked p -values (solid line) are always above the BH threshold (dashed line), so none of the alphas is significant when accounting for multiple testing. This finding is all the more notable given that the BH method is not as conservative as other multiple-testing adjustments (e.g., the Bonferroni correction or the Benjamini-Yekutieli method), meaning that the BH method is more prone to find significant greeniums.

Some readers might be concerned that our procedure does not detect any significant GMB factors because we inflate the number of tests by having many greenness measures and many ways of computing alphas. This concern would be justified if we used a method like Bonferroni, which indiscriminately lowers the p -value cutoff based on the number of tests. However, the BH method does not depend on the number of tests in such a simple way.

As seen in Figure 3(b), the BH method depends on the distribution of p -values given by the solid line. If we made ten copies of each p -value, the solid line would be unchanged, so the conclusions regarding significance would be unchanged. More broadly, considering many similar ways of computing alphas does *not* lead BH to find a lower fraction of significant greeniums if the distribution of p -values remains similar. Given that all our greenium estimates are ex ante about equally reasonable, their p -values should come from the same distribution, implying that our BH method does not become more stringent because of the large number of tests.

The reader might also wonder why our multiple-testing correction makes all greeniums insignificant when the asset-pricing literature finds that most equity factors survive multiple-

testing corrections (see, e.g., [Jensen et al., 2023](#)). In other words, why do we find a replication crisis when the literature on equity factors did not have one? To see why, simply consider the magnitude of the t -statistics. In asset pricing, most of the replicated t -statistics are above 2, frequently exceed 4, and are sometimes as high as 6 (see, e.g., [Chen, 2021](#)). In contrast, most replicated greenium t -statistics are close to zero and the largest is 3.01 (arising for the industry-neutral GMB factor based on MSCI’s weighted ESG score relative to the Fama-French three-factor model). This difference in magnitude arises because the asset pricing factors use many decades of data on potentially high-turnover strategies designed to deliver high returns, whereas the GMB factors use short samples for low-turnover strategies motivated by environmental rather than financial concerns.

In contrast to the largest absolute t -statistic of 3.01 in our scientific replication, the largest absolute published t -statistic in Figure 1 is -5.15 for a 13-year sample period (see Table 6 in [Bolton and Kacperczyk, 2023](#)). Our scientific replication does not produce this large t -statistic since we lag the emission data based on its release date, whereas the large t -statistic from the literature relies on using data before it was released, creating a look-ahead bias as documented in [Zhang \(2025\)](#). A t -statistic of -5.15 is not just outside our replication distribution, it is also outside what we expect given the short sample. To see why, note that a t -statistic corresponds to a corresponding trading strategy’s Sharpe ratio multiplied by $\sqrt{T}$, where T is the sample length (and where other exposures are hedged out, depending on the risk controls). Hence, a t -statistic of -5.15 over 13 years corresponds to an annual Sharpe ratio of around $5.15/\sqrt{13} = 1.43$, where we switch to a positive sign, that is, consider the corresponding brown-minus-green portfolio instead of GMB. A Sharpe ratio of 1.43 would make the strategy the best trading strategy among all the 153 US value-weighted capped factors considered in [Jensen et al. \(2023\)](#), where the median annual SR is around 0.3 and the largest is 1.06. In other words, we should not expect to find such a high t -statistic unless we expect the brown-minus-green portfolio to be the best factor in the literature by a wide margin, despite not being designed to maximize returns.⁸

⁸We note that the t -statistic of -5.15 comes from a regression rather than a long-short factor. Nevertheless, since regressions and portfolio sorts perform similar tests of the same issue, the t -statistic may

4 Out-of-Sample Evidence: Global Replication

So far, we have focused on US stocks, but it is also interesting to consider out-of-sample evidence in other countries. We next study realized return differences of global green and brown stocks outside the US. Figure 3(c) shows the distribution of alpha t -statistics for global ex-US GMB factors. Specifically, we compute a GMB factor for each country, greenness measure, and industry adjustment in the same way as the US factors. We then compute its alpha with respect to local risk models corresponding to the US Fama-French models: 1) no risk adjustment, 2) the local market, 3) the local market, size, and value, and 4) the local market, size, value, operating profitability, asset growth, and momentum.

As seen in Figure 3(c), the distribution of the global alphas' t -statistics is centered near zero and close to Normal. In other words, the distribution of the global greenium t -statistics almost looks the way it would look if all the greeniums were near zero plus random noise.

Figure 3(d) shows the cumulative distribution of the corresponding p -values. As seen in the figure, the ranked p -values (solid line) are always above the BH threshold (dashed line). Thus, as in the US, none of the alphas is significant when accounting for multiple testing. In fact, the ranked p -values are close to the 45-degree line, which indicates pure uncorrelated noise, echoing the nearly mean-zero Normal distribution of t -statistics.

In sum, looking in the US (as the literature) or abroad (out of sample), we do not find evidence of a significant greenium, as also seen in the merged US and ex-US global sample in Figure 1.

5 Evidence on Publication Bias

Relative to our scientific replication experiment, the distribution of published t -statistics in Figure 1 appears bimodal, with published t -statistics clustering at around -4 and $+2$. A

be expected to be of similar magnitude. Indeed, in a Fama-MacBeth regression of returns on characteristics, the regression coefficient can literally be seen as the average return of a trading strategy, where the t -statistic equals $SR \times \sqrt{T}$ (see, e.g., Pedersen (2015), Section 3.4), but this relation need not be exact in a panel regression and when standard errors are computed differently, as is the case in Bolton and Kacperczyk (2023).

bi-modal distribution is puzzling if the published estimates are interpreted as noisy measurements of the same underlying object—the greenium—since sampling variation alone would typically generate a unimodal distribution. As a case in point, our replicated distribution of t -statistics displays the expected unimodal, bell-shaped distribution shown in Figure 1. To formalize this comparison, we test whether the published and replicated t -statistics are drawn from the same distribution using a Kolmogorov-Smirnov test.

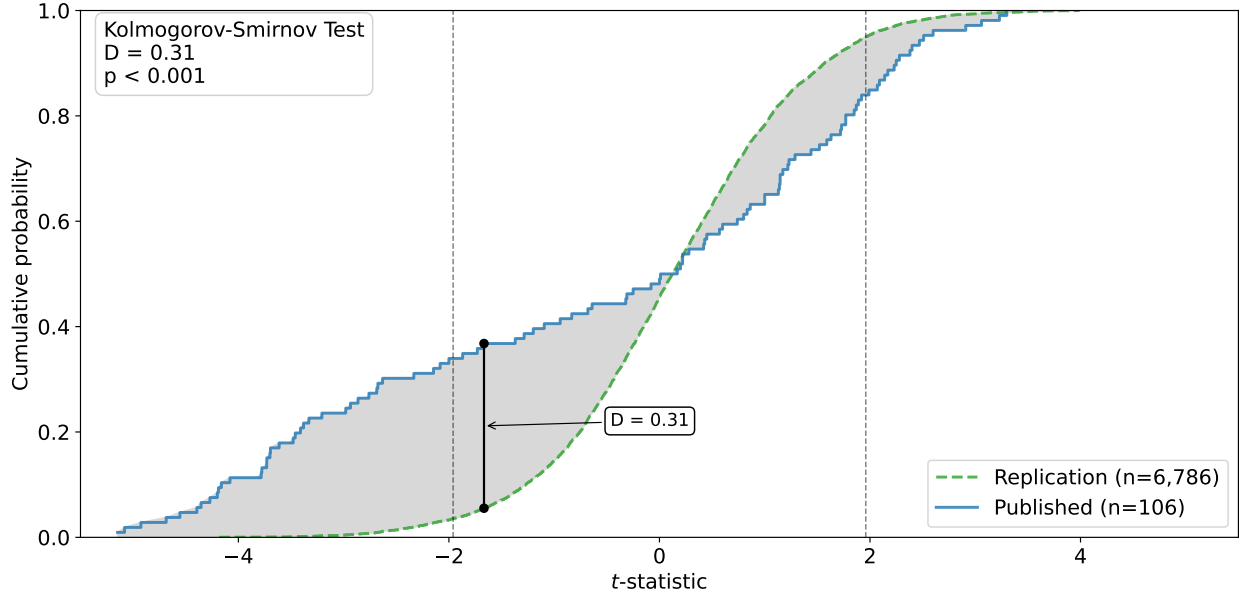
The Kolmogorov-Smirnov test statistic is the largest absolute difference between two empirical cumulative distribution functions (CDFs), in our case the CDFs of published and replicated t -statistics. Intuitively, if both samples truly come from the same distribution, their empirical CDFs should be close everywhere. A large test statistic indicates that the CDFs show substantial divergence.

Figure 4 plots the CDFs of replicated and published t -statistics. The figure also show the Kolmogorov-Smirnov test statistic, $D = 0.31$, meaning that the two CDFs differ by 0.31 at the shown point in the distribution. Statistically, this maximum difference is large enough to reject the null hypothesis that these samples come from the same underlying distribution (p -value < 0.001).⁹

Thus, the evidence is inconsistent with the hypothesis that the published test statistics are random samples from the replication distribution. Instead, the evidence is consistent with the hypothesis that a test is more likely to be published when the absolute value of the t -statistic is large. Our scientific replication experiment attempts to correct for any such bias by seeking to include the full set of ex ante plausible GMB factors, countries, and risk adjustments.

⁹We note that the large Kolmogorov-Smirnov test statistic is driven by the left mode in the distribution of published t -statistics, but it is not driven by the existence of t -statistics as low as -5.15 since $D = 0.31$ is achieved at a t -statistic around -1.7 as seen in Figure 4. In other words, D would be the same if all published t -statistics were winsorized below, for example at -2 .

Figure 4: Testing for Publication Bias via Difference in Distributions



The figure shows the cumulative distribution functions of published t -statistics (solid line) and t -statistics from our scientific replication (dashed line). The Kolmogorov-Smirnov test rejects that these distributions are the same (p -value < 0.001).

6 Why Realized Returns Don't Identify Greenium

Identifying the greenium using realized returns is difficult because the data cover little more than a single decade and because this sample simply has a too low signal-to-noise ratio. To see this problem more explicitly, we seek to estimate how much data is needed to identify the expected GMB return using realized returns.

To this end, note first that the realized annualized volatility of a GMB factor is 5.4% on average, with little variation across factors. Second, based on the forward-looking returns in [Eskildsen et al. \(2026\)](#), the largest magnitude of expected annualized returns among these factors is 0.78%.¹⁰ Hence, the predicted maximum annual Sharpe ratio is $-0.78/5.4 = -0.15$.

¹⁰Using their baseline greenium estimate of -39 bps per year per standard deviation increase in the green score, we predict an annual factor return of around $-39\text{bps} \times 2 = -78$ bps given that the GMB factor has a spread in green scores of 2 standard deviations.

Given a sample length of $T = 13.33$ years, the expected t -statistic is

$$t\text{-statistic} = \frac{E(r)}{\sigma/\sqrt{T}} = \text{SR} \times \sqrt{T} = -0.55. \quad (1)$$

Thus, the lack of statistical significance in realized returns is expected, even if a small greenium exists.

We can also consider how many years T it would take to get a t -statistic of 1.96 (the typical cut-off for significance at the 5% level):

$$T = \left(\frac{1.96}{\text{SR}} \right)^2 = 167 \text{ years}. \quad (2)$$

Naturally, since estimating the overall equity premium is known to require a long sample (Merton, 1980), estimating its green component requires an even longer sample.

As another way to illustrate the problem, we can ask what SR is needed to reach a t -statistic of at least 1.96 in Equation (1). The answer is that the SR must be higher than 0.54. In other words, the greenium must have a larger SR than the equity market and the majority of equity factors in the literature. For example, the median SR is 0.3 among the 153 US value-weighted capped factors considered in Jensen et al. (2023), and 0.54 lies at the top 79th percentile among factors designed to be profitable, whereas the greenium is not designed to maximize returns.

Estimating the greenium is made even more difficult by the fact that the sample period is a time of rising green investor concerns. Such rising concerns may lead to a repricing, where green stocks temporarily outperform as the forward-looking expected returns fall. In other words, during a time of repricing, expected returns and realized returns can move in opposite directions.

As a case in point, the literature uses both realized out- and underperformance of green stocks to support the existence of a negative greenium. Some argue that realized underperformance can be seen as an estimate of the unconditional greenium (e.g., Bolton and Kacperczyk, 2021), while others argue that the realized outperformance is a sign of repricing

(Pástor et al., 2022).

In summary, the noise in realized returns from repricing of the greenium, shocks to cashflows, and the short sample period mean that the greenium cannot be robustly identified from realized returns with the currently available data.

7 Conclusion

This paper studies the returns to green investing, an issue that has attracted enormous academic and policy interest, yet produced strikingly conflicting answers. By systematically replicating the literature across greenness measures, portfolio constructions, risk models, and countries, we find that realized return differences between green and brown stocks are statistically fragile. Once we account for multiple testing, there is no statistically reliable evidence of a greenium based on realized returns. Our findings clarify how conflicting results can arise even in leading academic journals and why practitioners remain divided on whether green investing earns a premium.

The deeper lesson is not that environmental preferences or risks are unimportant, but that realized equity returns are a blunt instrument for measuring them. When expected return differences are small relative to equity volatility and limited sample spans, empirical results are inevitably sensitive to specification choices. In such settings, disagreement between studies is a consequence of noisy data, potentially coupled with publication bias.

Understanding how climate concerns affect asset prices requires moving beyond realized returns toward economic theory, forward-looking return measures, price reactions to climate-related news and regulatory shocks, and evidence from investor behavior.

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A Internet Appendix

A.1 Lagging conventions and sample period

Lag conventions

Following [Eskildsen et al. \(2026\)](#), we adopt the following adjustments to ensure that the greenness data are publicly available to investors, free from look-ahead bias and staleness:

- Trucost data are assumed to be available by the end of the month in which emissions are estimated by Trucost or disclosed by the firm, following [Pedersen et al. \(2021\)](#) and [Zhang \(2025\)](#). We use Trucost data for up to three years following initial availability, subject to a lag of at most five years relative to the corresponding firm reporting date.
- EPA TRI data are assumed to be available in September of the year after reporting, following [Hsu et al. \(2023\)](#), and are used for up to one year.
- When scaling a firm’s Trucost or EPA emissions by sales or assets, we use sales or assets from the most recent fiscal period end in the same calendar year (typically December). Accounting data are assumed to be available four months after the fiscal period end, following [Jensen et al. \(2023\)](#).
- MSCI and Sustainalytics data are used for up to one year.

Sample period

The analysis uses the full time series available for each greenness measure. Applying the lag conventions above yields the following sample periods:

- Trucost data are available from May 2009.
- MSCI data are available from January 2007, except for the four Environmental sub-scores (Climate Change, Natural Resources, Pollution & Waste, and Environmental Opportunities), which are only available from January 2013.
- Sustainalytics data are available from August 2009.
- EPA TRI data are available from September 1992.

- For the out-of-sample portfolio sorts, the greenness data are lagged by one month. As an example, MSCI data for August 2010 are used for portfolio sorts in September 2010.
- All analyses are conducted through December 2022.

A.2 Published t -statistics

Table A1 shows the t -statistics used to construct the histogram of published t -statistics in Panel a) of Figure 1. We collect t -statistics for papers on the realized performance of green versus brown stocks published in the Journal of Financial Economics (JFE), Journal of Finance (JF), Review of Financial Studies (RFS), and the Review of Finance (RoF). For each paper, we collect the t -statistics from what we consider the main tables of the paper and sign them such that a positive t -statistic indicates green outperformance over brown and a negative t -statistic indicates brown outperformance over green.

Table A1: t -statistics published in the literature

Paper	Sample	t -stat.	Measure	Industry adj.	Risk model	Reference	Journal
Bolton and Kacperczyk (2021)	US	-2.33	LOG(S2TOT)	No	Excess returns	Table 8	JFE
Bolton and Kacperczyk (2021)	US	-2.93	LOG(S3TOT)	No	Excess returns	Table 8	JFE
Bolton and Kacperczyk (2021)	US	-4.56	LOG(S1TOT)	Yes	Excess returns	Table 8	JFE
Bolton and Kacperczyk (2021)	US	-3.48	LOG(S2TOT)	Yes	Excess returns	Table 8	JFE
Bolton and Kacperczyk (2021)	US	-4.39	LOG(S3TOT)	Yes	Excess returns	Table 8	JFE
Bolton and Kacperczyk (2021)	US	-4.19	Delta S1TOT	No	Excess returns	Table 8	JFE
Bolton and Kacperczyk (2021)	US	-2.76	Delta S2TOT	No	Excess returns	Table 8	JFE
Bolton and Kacperczyk (2021)	US	-3.78	Delta S3TOT	No	Excess returns	Table 8	JFE
Bolton and Kacperczyk (2021)	US	-4.35	Delta S1TOT	Yes	Excess returns	Table 8	JFE
Bolton and Kacperczyk (2021)	US	-2.68	Delta S2TOT	Yes	Excess returns	Table 8	JFE
Bolton and Kacperczyk (2021)	US	-3.78	Delta S3TOT	Yes	Excess returns	Table 8	JFE
Bolton and Kacperczyk (2021)	US	0.83	S1INT (Sales)	No	Excess returns	Table 8	JFE
Bolton and Kacperczyk (2021)	US	-1.20	S2INT (Sales)	No	Excess returns	Table 8	JFE
Bolton and Kacperczyk (2021)	US	-1.67	S3INT (Sales)	No	Excess returns	Table 8	JFE
Bolton and Kacperczyk (2021)	US	-0.83	S1INT (Sales)	Yes	Excess returns	Table 8	JFE
Bolton and Kacperczyk (2021)	US	-1.09	S2INT (Sales)	Yes	Excess returns	Table 8	JFE
Bolton and Kacperczyk (2021)	US	-0.64	S3INT (Sales)	Yes	Excess returns	Table 8	JFE
Bolton and Kacperczyk (2023)	US	-3.38	LOG(S1TOT)	No	Excess returns	Table 5	JF
Bolton and Kacperczyk (2023)	US	-2.08	LOG(S2TOT)	Yes	Excess returns	Table 5	JF
Bolton and Kacperczyk (2023)	US	-2.86	LOG(S3TOT)	Yes	Excess returns	Table 5	JF
Bolton and Kacperczyk (2023)	US	-4.27	Delta S1TOT	Yes	Excess returns	Table 5	JF
Bolton and Kacperczyk (2023)	US	-2.15	Delta S2TOT	Yes	Excess returns	Table 5	JF
Bolton and Kacperczyk (2023)	US	-2.69	Delta S3TOT	Yes	Excess returns	Table 5	JF
Bolton and Kacperczyk (2023)	Global	-1.29	LOG(S1TOT)	No	Excess returns	Table 6	JF
Bolton and Kacperczyk (2023)	Global	-3.21	LOG(S2TOT)	No	Excess returns	Table 6	JF
Bolton and Kacperczyk (2023)	Global	-3.61	LOG(S3TOT)	No	Excess returns	Table 6	JF
Bolton and Kacperczyk (2023)	Global	-4.20	LOG(S1TOT)	Yes	Excess returns	Table 6	JF
Bolton and Kacperczyk (2023)	Global	-4.93	LOG(S2TOT)	Yes	Excess returns	Table 6	JF
Bolton and Kacperczyk (2023)	Global	-4.69	LOG(S3TOT)	Yes	Excess returns	Table 6	JF
Bolton and Kacperczyk (2023)	Global	-5.08	Delta S1TOT	No	Excess returns	Table 6	JF
Bolton and Kacperczyk (2023)	Global	-3.73	Delta S2TOT	No	Excess returns	Table 6	JF
Bolton and Kacperczyk (2023)	Global	-4.16	Delta S3TOT	No	Excess returns	Table 6	JF
Bolton and Kacperczyk (2023)	Global	-5.15	Delta S1TOT	Yes	Excess returns	Table 6	JF
Bolton and Kacperczyk (2023)	Global	-3.70	Delta S2TOT	Yes	Excess returns	Table 6	JF
Bolton and Kacperczyk (2023)	Global	-4.08	Delta S3TOT	Yes	Excess returns	Table 6	JF
Zhang (2025)	US	2.47	S1INT (Sales)	No	Excess returns	Table 4	JF
Zhang (2025)	US	1.87	S2INT (Sales)	No	Excess returns	Table 4	JF
Zhang (2025)	US	2.51	S1INT (Sales)	No	FF5 + MOM alphas	Table 4	JF
Zhang (2025)	US	2.40	S2INT (Sales)	No	FF5 + MOM alphas	Table 4	JF
Zhang (2025)	US	-1.37	Delta S1TOT	No	Excess returns	Table 4	JF
Zhang (2025)	US	0.17	Delta S2TOT	No	Excess returns	Table 4	JF
Zhang (2025)	US	-0.68	Delta S3TOT	No	FF5 + MOM alphas	Table 4	JF
Zhang (2025)	US	0.57	Delta S2TOT	No	FF5 + MOM alphas	Table 4	JF
Zhang (2025)	US	1.77	LOG(S1TOT)	No	Excess returns	Table 4	JF
Zhang (2025)	US	0.42	LOG(S2TOT)	No	Excess returns	Table 4	JF
Zhang (2025)	US	3.30	LOG(S1TOT)	No	FF5 + MOM alphas	Table 4	JF
Zhang (2025)	US	2.17	LOG(S2TOT)	No	FF5 + MOM alphas	Table 4	JF
Zhang (2025)	Global	0.20	S1INT (Sales)	No	Excess returns	Table 7	JF
Zhang (2025)	Global	-0.08	S2INT (Sales)	No	Excess returns	Table 7	JF
Zhang (2025)	Global	0.74	S1INT (Sales)	No	FF5 + MOM alphas	Table 7	JF
Zhang (2025)	Global	0.43	S2INT (Sales)	No	FF5 + MOM alphas	Table 7	JF
Zhang (2025)	Global	1.13	Delta S1TOT	No	Excess returns	Table 7	JF
Zhang (2025)	Global	-0.25	Delta S2TOT	No	Excess returns	Table 7	JF
Zhang (2025)	Global	1.22	Delta S1TOT	No	FF5 + MOM alphas	Table 7	JF
Zhang (2025)	Global	0.00	Delta S2TOT	No	FF5 + MOM alphas	Table 7	JF
Zhang (2025)	Global	1.77	LOG(S1TOT)	No	Excess returns	Table 7	JF
Zhang (2025)	Global	3.06	LOG(S2TOT)	No	Excess returns	Table 7	JF
Zhang (2025)	Global	1.44	LOG(S1TOT)	No	FF5 + MOM alphas	Table 7	JF
Zhang (2025)	Global	2.07	LOG(S2TOT)	No	FF5 + MOM alphas	Table 7	JF
Aswani et al. (2024)	US	1.17	LOG(S1TOT)	No	Excess returns	Table 5	RoF
Aswani et al. (2024)	US	1.15	LOG(S2TOT)	No	Excess returns	Table 5	RoF
Aswani et al. (2024)	US	1.00	LOG(S3TOT)	No	Excess returns	Table 5	RoF
Aswani et al. (2024)	US	1.23	LOG(S1TOT)	Yes	Excess returns	Table 5	RoF
Aswani et al. (2024)	US	1.15	LOG(S2TOT)	Yes	Excess returns	Table 5	RoF
Aswani et al. (2024)	US	0.86	LOG(S3TOT)	Yes	Excess returns	Table 5	RoF
Aswani et al. (2024)	US	2.60	S1INT (Sales)	Yes	Excess returns	Table 8	RoF
Aswani et al. (2024)	US	1.29	S2INT (Sales)	Yes	Excess returns	Table 8	RoF
Aswani et al. (2024)	US	-0.94	S3INT (Sales)	Yes	Excess returns	Table 8	RoF
Pástor et al. (2022)	US	3.23	Greenness (PST)	No	Excess returns	Table 3	JFE
Pástor et al. (2022)	US	2.91	Greenness (PST)	No	CAPM alphas	Table 3	JFE
Pástor et al. (2022)	US	2.23	Greenness (PST)	No	FF3 alphas	Table 3	JFE
Pástor et al. (2022)	US	2.14	Greenness (PST)	No	FF4 alphas	Table 3	JFE
Pástor et al. (2022)	US	2.25	Greenness (PST)	No	FF3 + LIQ	Table 3	JFE
Pástor et al. (2022)	US	2.38	Greenness (PST)	No	FF5 alphas	Table 3	JFE
Pástor et al. (2022)	US	2.28	Greenness (PST)	No	CAPM + controls	Table 3	JFE
Pástor et al. (2022)	US	1.99	Greenness (PST)	No	CAPM + controls	Table 3	JFE
Hsu et al. (2023)	US	-1.73	TPWINT (Sales)	Yes	Excess returns	Table 2	JF
Hsu et al. (2023)	US	-3.46	TPWINT (Assets)	Yes	Excess returns	Table 2	JF
Hsu et al. (2023)	US	-2.00	TPWINT (PPENT)	Yes	Excess returns	Table 2	JF
Hsu et al. (2023)	US	-2.63	TPWINT (ME)	Yes	Excess returns	Table 2	JF
Hsu et al. (2023)	US	-3.41	TPWINT (Assets)	Yes	CAPM alphas	Table 4	JF
Hsu et al. (2023)	US	-3.73	TPWINT (Assets)	Yes	FF3 alphas	Table 4	JF
Hsu et al. (2023)	US	-3.33	TPWINT (Assets)	Yes	FF4 alphas	Table 4	JF
Hsu et al. (2023)	US	-2.98	TPWINT (Assets)	Yes	FF5 alphas	Table 4	JF
Hsu et al. (2023)	US	-3.70	TPWINT (Assets)	Yes	q-factor alphas	Table 4	JF
Pedersen et al. (2021)	US	0.28	Ind-adj. ESG score	Yes	Excess returns	Table 4 Panel A	JFE
Pedersen et al. (2021)	US	1.59	S1+S2INT (Sales)	No	Excess returns	Table 4 Panel A	JFE
Pedersen et al. (2021)	US	0.01	Ind-adj. ESG score	Yes	Excess returns	Table 4 Panel B	JFE
Pedersen et al. (2021)	US	1.89	S1+S2INT (Sales)	No	Excess returns	Table 4 Panel B	JFE
Pedersen et al. (2021)	US	1.00	Ind-adj. ESG score	Yes	CAPM alphas	Table 4 Panel A	JFE
Pedersen et al. (2021)	US	2.09	S1+S2INT (Sales)	No	CAPM alphas	Table 4 Panel A	JFE
Pedersen et al. (2021)	US	0.80	Ind-adj. ESG score	Yes	CAPM alphas	Table 4 Panel B	JFE
Pedersen et al. (2021)	US	1.52	S1+S2INT (Sales)	No	CAPM alphas	Table 4 Panel B	JFE
Pedersen et al. (2021)	US	0.60	Ind-adj. ESG score	Yes	FF3 alphas	Table 4 Panel A	JFE
Pedersen et al. (2021)	US	1.63	S1+S2INT (Sales)	No	FF3 alphas	Table 4 Panel A	JFE
Pedersen et al. (2021)	US	0.45	Ind-adj. ESG score	Yes	FF3 alphas	Table 4 Panel B	JFE
Pedersen et al. (2021)	US	1.14	S1+S2INT (Sales)	No	FF3 alphas	Table 4 Panel B	JFE
Pedersen et al. (2021)	US	0.22	Ind-adj. ESG score	Yes	FF5 alphas	Table 4 Panel A	JFE
Pedersen et al. (2021)	US	1.92	S1+S2INT (Sales)	No	FF5 alphas	Table 4 Panel A	JFE
Pedersen et al. (2021)	US	-0.31	Ind-adj. ESG score	Yes	FF5 alphas	Table 4 Panel B	JFE
Pedersen et al. (2021)	US	1.85	S1+S2INT (Sales)	No	FF5 alphas	Table 4 Panel B	JFE
Pedersen et al. (2021)	US	0.22	Ind-adj. ESG score	Yes	FF5 + MOM alphas	Table 4 Panel A	JFE
Pedersen et al. (2021)	US	1.73	S1+S2INT (Sales)	No	FF5 + MOM alphas	Table 4 Panel A	JFE
Pedersen et al. (2021)	US	-0.32	Ind-adj. ESG score	Yes	FF5 + MOM alphas	Table 4 Panel B	JFE
Pedersen et al. (2021)	US	1.72	S1+S2INT (Sales)	No	FF5 + MOM alphas	Table 4 Panel B	JFE

A.3 Replicated t -statistics

Tables A2 and A3 show the t -statistics of the 46 US GMB factors' returns, constructed as either industry-neutral or industry-agnostic portfolios based on all greenness measures, and then testing the alpha with respect to all the different factor models. The tables show that the vast majority of t -statistics are close to zero. Further, for the small minority of significant greeniums, the greenium for the same greenness measure is insignificant when using another risk model or another industry adjustment.

Table A2: Alphas and t -statistics of industry-agnostic GMB equity factors

	SR	r	$t(r)$	α^{CAPM}	$t(\alpha^{CAPM})$	α^{FF3}	$t(\alpha^{FF3})$	α^{FF6}	$t(\alpha^{FF6})$	α^{q5}	$t(\alpha^{q5})$
LOG(S1TOT)	0.11	0.07	0.37	-0.05	-0.27	-0.01	-0.07	0.29	2.87	0.17	1.36
LOG(S1+2TOT)	0.06	0.05	0.21	-0.06	-0.25	-0.00	-0.03	0.31	2.97	0.22	1.59
LOG(S1+2+3TOT)	-0.06	-0.05	-0.21	-0.13	-0.51	-0.08	-0.47	0.24	2.45	0.18	1.34
S1INT (Sales)	0.24	0.13	0.84	0.11	0.67	0.10	0.66	0.28	2.22	0.13	0.97
S1+2INT (Sales)	0.25	0.13	0.90	0.10	0.63	0.09	0.65	0.29	2.48	0.14	1.04
S1+2+3INT (Sales)	0.15	0.09	0.52	0.08	0.46	0.07	0.50	0.30	2.53	0.18	1.31
S1INT (Assets)	0.20	0.11	0.69	0.07	0.39	0.06	0.39	0.28	2.32	0.13	0.90
S1+2INT (Assets)	0.17	0.09	0.59	0.04	0.24	0.04	0.25	0.29	2.80	0.15	1.05
S1+2+3INT (Assets)	-0.02	-0.01	-0.05	-0.07	-0.43	-0.06	-0.44	0.22	2.22	0.09	0.72
Ind.-adj. ESG score	0.43	0.16	1.66	0.23	2.78	0.20	2.94	0.17	2.29	0.08	1.14
Weighted ESG score	0.38	0.16	1.46	0.23	2.18	0.19	2.03	0.13	1.32	0.00	0.05
Environment score	0.25	0.13	0.87	0.21	1.29	0.12	1.09	0.18	1.53	0.08	0.69
Greenness (PST)	0.03	0.02	0.10	0.02	0.12	-0.02	-0.08	0.24	1.28	-0.02	-0.10
E climate score	0.49	0.32	1.36	0.41	1.57	0.30	2.73	0.27	2.15	0.23	1.48
E nat. res. score	0.40	0.22	1.25	0.35	1.82	0.26	1.82	0.25	1.86	0.19	1.36
E waste score	0.22	0.19	0.75	0.49	1.71	0.40	2.13	0.34	1.77	0.27	1.37
E env. opps. score	0.07	0.04	0.23	0.13	0.77	0.07	0.45	0.04	0.31	-0.02	-0.16
Total ESG score	-0.01	-0.00	-0.02	0.12	1.03	0.10	1.06	-0.02	-0.24	-0.09	-0.94
Environmental score	0.31	0.15	1.09	0.27	2.02	0.24	2.29	0.10	1.03	0.05	0.52
TRI (Sales)	0.12	0.06	0.67	0.05	0.47	0.10	1.08	0.19	1.96	0.10	1.02
TPWINT (Sales)	-0.01	-0.00	-0.04	0.08	1.10	0.11	1.50	0.13	1.57	0.09	1.13
TRI (Assets)	0.13	0.07	0.66	0.07	0.70	0.13	1.44	0.22	2.17	0.10	1.02
TPWINT (Assets)	0.02	0.01	0.13	0.13	1.53	0.16	2.05	0.18	2.04	0.12	1.31

The table shows Sharpe ratios (SR) as well as alphas and their corresponding t -statistics for 23 green-minus-brown (GMB) US equity factors. The 23 factors are constructed as the return difference of a portfolio that goes long the top tercile of stocks based on a greenness measure and short the bottom tercile. Portfolio returns are value-weighted capped stock returns with a cap on market capitalization at the NYSE 80th percentile. The 23 greenness measures are the 23 individual greenness measures from Table 1. We compute alphas with respect to five models: i) no risk adjustment (excess returns r), ii) the CAPM, iii) the Fama-French three-factor model, iv) the Fama-French five-factor model augmented by momentum, and v) the $q5$ -factor model. Standard errors are Newey and West (1987) adjusted with three lags.

Table A3: Alphas and t -statistics of industry-neutral GMB equity factors

	SR	r	$t(r)$	α^{CAPM}	$t(\alpha^{CAPM})$	α^{FF3}	$t(\alpha^{FF3})$	α^{FF6}	$t(\alpha^{FF6})$	α^{q5}	$t(\alpha^{q5})$
LOG(S1TOT)	0.00	0.00	0.02	0.00	0.01	0.02	0.24	0.05	0.69	0.11	1.32
LOG(S1+2TOT)	-0.12	-0.06	-0.41	-0.07	-0.40	-0.04	-0.45	0.00	0.01	0.07	0.70
LOG(S1+2+3TOT)	-0.18	-0.10	-0.60	-0.08	-0.43	-0.04	-0.47	-0.01	-0.06	0.09	0.83
S1INT (Sales)	0.23	0.06	0.85	0.07	0.89	0.05	0.78	0.04	0.64	0.04	0.54
S1+2INT (Sales)	0.20	0.05	0.76	0.08	1.06	0.07	1.27	0.04	0.81	0.06	0.93
S1+2+3INT (Sales)	0.15	0.04	0.62	0.06	0.71	0.05	1.01	0.06	1.18	0.07	1.28
S1INT (Assets)	0.06	0.02	0.20	-0.02	-0.25	-0.04	-0.52	0.00	0.03	-0.00	-0.05
S1+2INT (Assets)	-0.07	-0.02	-0.26	-0.06	-0.73	-0.08	-1.21	-0.05	-0.69	-0.02	-0.29
S1+2+3INT (Assets)	-0.12	-0.03	-0.48	-0.09	-1.32	-0.10	-1.62	-0.04	-0.68	-0.02	-0.28
Ind.-adj. ESG score	0.29	0.09	1.02	0.15	2.00	0.14	2.19	0.09	1.55	0.04	0.58
Weighted ESG score	0.45	0.15	1.60	0.22	2.82	0.20	3.01	0.16	2.29	0.10	1.48
Environment score	0.27	0.09	1.03	0.14	1.55	0.11	1.50	0.09	1.09	0.05	0.68
Greenness (PST)	0.20	0.07	0.84	0.11	1.30	0.08	1.19	0.08	1.25	0.06	0.92
E climate score	0.54	0.22	1.55	0.31	2.18	0.21	2.57	0.17	2.33	0.10	1.43
E nat. res. score	0.15	0.07	0.48	0.23	1.90	0.15	1.70	0.11	1.26	0.07	0.73
E waste score	0.22	0.09	0.80	0.19	1.60	0.14	1.34	0.08	0.79	0.06	0.58
E env. opps. score	0.11	0.04	0.44	0.04	0.40	0.00	0.01	0.04	0.49	0.02	0.22
Total ESG score	0.04	0.01	0.17	0.05	0.61	0.04	0.64	-0.00	-0.05	-0.08	-1.23
Environmental score	0.15	0.05	0.61	0.08	0.97	0.06	0.91	0.02	0.27	-0.05	-0.85
TRI (Sales)	0.00	0.00	0.01	0.09	1.19	0.11	1.59	0.13	1.67	0.11	1.37
TPWINT (Sales)	-0.13	-0.05	-0.72	0.02	0.35	0.03	0.51	0.06	0.85	0.07	0.90
TRI (Assets)	0.01	0.00	0.06	0.07	0.97	0.10	1.37	0.12	1.57	0.10	1.24
TPWINT (Assets)	-0.20	-0.08	-1.13	-0.01	-0.17	0.00	0.03	0.06	0.96	0.07	0.98

The table is similar to Table A2, but shows results for industry-neutral GMB US equity factors.

A.4 Pure vs. scientific replication

Our results in Tables A2 and A3 present our “scientific replication” of the literature, meaning that we examine the results from the literature using a common framework for all greenness measures, rather than following each paper’s specific steps using exactly the same sample as in “pure replication” or “reproduction.”

The advantage of scientific replication is that it shows the robustness of the literature’s results to (a) extending the sample over time and countries; (b) considering different risk models; and (c) a standard framework (not tailored to finding a significant effect for a specific measure); and (d) a unified framework, which makes the various greenium estimates comparable.

That said, it is interesting to relate our results to those in the literature. To be clear, we are not saying that the literature’s results cannot be reproduced if following each paper’s exact method. To the contrary, we now show that several results from the literature can be reproduced. Our point is that these results may be special relative to our full replication

distribution, that is, relying on a specific greenness measure, time period, and methodology.

Aswani et al. (2024); Zhang (2025) already discuss replication issues with Bolton and Kacperczyk (2021, 2023), so we consider instead a pure replication of some of the other main papers in the literature.

Reproducing and extending PST. We first consider a reproduction of Pástor et al. (2022) (henceforth “PST”). Columns (1)–(3) of Table A4 reproduce Columns (1)–(3) of Table 3 in PST, who form a GMB factor based on a transformation of MSCI’s ESG scores, with risk controls being, respectively no control, the market, and Fama-French three-factor model. These results nearly reproduce those in PST. For instance, they find a 65 basis points monthly mean excess return of their green-minus-brown (GMB) factor, whereas we find a 71 basis points monthly excess return.

Column (4) expands the risk control to include the Fama-French five-factor model augmented with momentum (as we use elsewhere), which does not affect the results much.

Columns (5)–(8) of Table A4 extend the sample period forward by two years until December 2022. The GMB factor’s excess returns and alphas become insignificant in this extended sample.

In sum, this analysis shows that PST can be reproduced, so the insignificant findings in our scientific replication in Table A2 reflects that all our analysis is performed with the extended sample. Likewise, the insignificant findings in Table A3, which uses industry-neutral factors, is consistent with the findings of PST, who also report insignificant results for an industry-neutral factor (see their Figure 8).

Further, PST emphasize that realized returns of the GMB factor need to be purged from unexpected shocks to climate concerns and cash-flow news. Once purged, the constant in a regression of the GMB factor’s return on climate change concern shocks and cash-flow news is a cleaner estimate of the greenium.

Columns (1) and (2) of Table A5 replicate Columns (1) and (2) of Table 4 in PST. As in the original paper, we find a significant effect of changes in climate concerns on the GMB factor. However, this effect largely turns insignificant when we extend the sample forward by two years in Columns (3) and (4), and backward until February 2007 in Columns (5) and (6). Columns (7)–(12) show similar results with Fama-French three-factor alphas as opposed to excess returns of the GMB factor as the dependent variables. In any case, the key quantity of interest—the greenium as measured by the constant—is not statistically different from zero in all specifications, consistent with PST’s original results.

Finally, Figure A1 shows the performance of the original GMB factor that is not industry-adjusted over time and compares with PST’s Figure 8. The figure also shows the climate change concerns index of Ardia et al. (2022) used by PST. While the GMB factor return reverses after PST’s sample period ended in December 2020, climate change concerns continued to increase until 2022.

Reproducing and extending Zhang. The second paper we reproduce is Zhang (2025). Zhang (2025) focuses on industry-agnostic GMB portfolios sorted on carbon intensity (appropriately lagged scope 1 or scope 2 carbon emissions scaled by year-end sales) from June 2009 to December 2021. Figure A2 reproduces Zhang (2025)’s GMB factor’s return based on scope 1 carbon emissions scaled by sales as in her Figure 2. Once again, we can reproduce her results, but when we extend the sample forward by one year, the GMB factor’s returns partially reverse. This explains that our scientific replication in Table A2 shows no evidence of significant excess returns for a GMB factor formed based on scope 1 carbon emissions scaled by sales (S1INT (Sales)). The monthly mean excess return for this factor is only 13 basis points.

There is evidence of significant Fama-French six-factor alphas in our Table A2—the factor model considered by Zhang (2025)—for portfolios formed based on scope 1 carbon emissions scaled by sales. However, these are not robust to the factor model (i.e., there are no significant CAPM, Fama-French three-factor, or $q5$ -factor alphas). Moreover, these alphas, which are significant using conventional significance tests, do not survive a multiple testing adjustment. Finally, there is no evidence of significant excess returns or alphas of industry-neutral factors based on carbon emissions scaled by scales as shown in our Table A3, which is a result consistent with Zhang (2025)’s Table 6.

As another example, Hsu et al. (2023) find that an industry-neutral GMB portfolio based on toxic emission intensity generates a significant annual return of -4.42% from 1991 to 2016. Our scientific replication shows that a long-short portfolio sorted on toxic emission intensity (TPWINT (Assets)) has an insignificant excess return. There are several methodological differences, including that this paper uses quintile as opposed to tercile sorts for portfolio construction and that their industry-adjustment is based on SIC codes only from CRSP while we use the standard Fama-French approach.¹¹ In any case, the EPA metric used by

¹¹We follow Fama and French and use SIC codes from Compustat and only supplement with CRSP whenever these are not available, see https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/Data_Library/det_49_ind_port.html. Kahle and Walkling (1996) find substantial divergence between SIC codes from CRSP and Compustat with more than 80% disagreement at the 4-digit level used to generate FF49

Table A4: PST’s GMB performance

	Exc. Ret. (1)	CAPM (2)	FF3 (3)	FF6 (4)	Exc. Ret. (5)	CAPM (6)	FF3 (7)	FF6 (8)
Constant	0.71*** (0.23)	0.81*** (0.29)	0.56** (0.26)	0.54** (0.25)	0.24 (0.24)	0.27 (0.28)	0.22 (0.21)	0.31 (0.21)
Mkt-RF		-0.08 (0.08)	0.01 (0.06)	0.06 (0.07)		-0.02 (0.08)	0.02 (0.05)	0.04 (0.06)
SMB			-0.18* (0.11)	-0.27** (0.13)			-0.14 (0.09)	-0.29** (0.12)
HML			-0.25** (0.10)	-0.14 (0.15)			-0.39*** (0.06)	-0.22** (0.09)
RMW				-0.36** (0.17)				-0.33** (0.14)
CMA				-0.03 (0.23)				-0.26* (0.14)
MOM				0.12 (0.08)				0.03 (0.07)
Adj. R^2	-0.00	0.01	0.18	0.23	-0.00	-0.01	0.32	0.38
N	98	98	98	98	122	122	122	122

The table shows returns of a green-minus-brown factor regressed on different factor models. Columns (1)–(3) (purely) reproduce Columns (1)–(3) of Table 3 in [Pástor et al. \(2022\)](#). Column (4) uses the Fama-French five-factor model augmented with momentum over the same sample period. Columns (5)–(8) extend the sample period from December 2020 until December 2022. Standard errors (in parentheses) are heteroskedasticity robust. *, **, and *** denote 10%, 5%, and 1% significance levels, respectively, for the null hypothesis of a zero coefficient. N refers to the total number of observations.

[Hsu et al. \(2023\)](#) does not distinguish between more or less dangerous chemicals and does not even distinguish between whether they have been recycled versus released (into the air, water, or ground) even though these data exist. For instance, our TRINT variable only considers released toxic waste, while their TPWINT variable does not distinguish one pound of cyanide released into the water versus one pound of paint waste that is recycled.¹²

Finally, the level of carbon emissions—as opposed to carbon intensity (i.e., emissions scaled by sales or assets)—has received significant attention in the literature (see, e.g., [Bolton and Kacperczyk, 2021, 2023](#)). However, once again, Tables [A2](#) and [A3](#) show that there is little evidence for significant mean excess returns or alphas of portfolios sorted on the industry classifications.

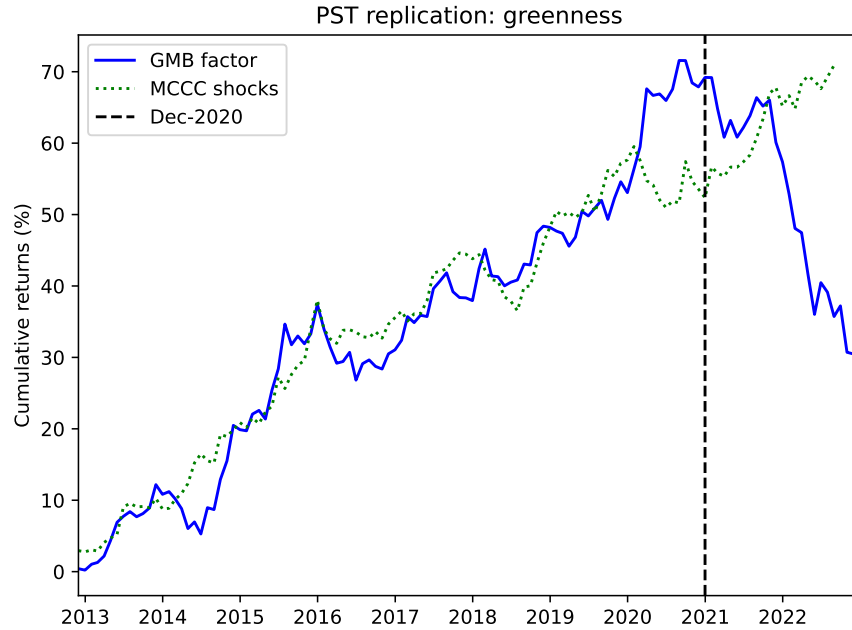
¹²For completeness, [Hsu et al. \(2023\)](#) consider toxicity in their Internet Appendix using a county-level mortality model (as opposed to simply scaling by EPA’s toxicity estimates).

Table A5: PST’s GMB returns, changes in climate concerns, and cash-flow news

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Constant	0.19 (0.28)	0.06 (0.28)	0.23 (0.29)	0.04 (0.25)	0.09 (0.21)	0.05 (0.20)	0.01 (0.25)	-0.08 (0.25)	0.06 (0.21)	0.00 (0.20)	0.01 (0.20)	0.00 (0.19)
Δ Climate concerns (same month)	2.16* (1.14)	2.14* (1.12)	1.11 (0.90)	1.44* (0.77)	0.54 (0.67)	0.76 (0.65)	2.40** (0.96)	2.40** (0.94)	1.37** (0.66)	1.50** (0.63)	0.59 (0.62)	0.79 (0.62)
Δ Climate concerns (prev. month)	2.21* (1.13)	1.82* (1.02)	-0.37 (1.00)	-0.54 (0.92)	-0.41 (0.75)	-0.31 (0.74)	1.95** (0.89)	1.68** (0.83)	0.35 (0.69)	0.31 (0.68)	-0.21 (0.66)	-0.11 (0.67)
Earnings announcement returns		0.35* (0.21)		0.43*** (0.14)		0.25** (0.11)		0.29 (0.18)		0.23** (0.10)		0.21* (0.11)
Δ Earnings forecasts		0.04 (0.14)		-0.07 (0.13)		-0.27*** (0.10)		-0.04 (0.13)		-0.13 (0.10)		-0.30*** (0.10)
Adj. R^2	0.12	0.18	0.00	0.11	-0.01	0.06	0.14	0.17	0.02	0.06	-0.01	0.07
N	68	68	118	118	187	187	68	68	118	118	187	187

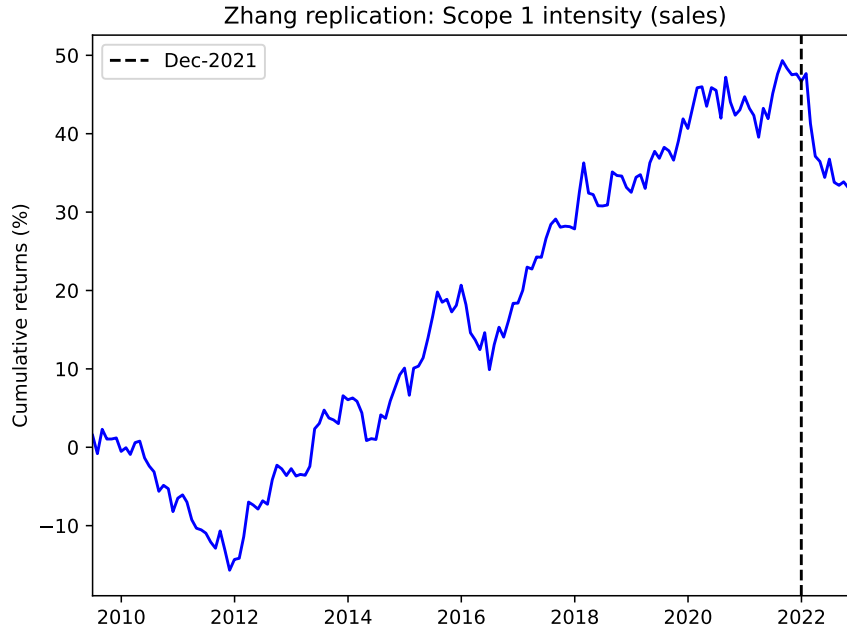
The table shows regressions of a green-minus-brown (GMB) factor’s excess returns or Fama-French three-factor alpha on a constant, contemporaneous and lagged changes in climate concerns, and two earnings measures. Columns (1)–(6) have excess return as the dependent variable, whereas Columns (7)–(12) have alphas as the dependent variable. The GMB factor is constructed using the greenness measure from [Pástor et al. \(2022\)](#), and the earnings announcement return and Δ earnings forecast factors come from [Chen and Zimmermann \(2022\)](#). Changes in climate concerns are constructed as in [Pástor et al. \(2022\)](#). Specifications (1), (2), (7), and (8) use a sample period from 2012–2018 and reproduce specifications (1)–(4) in Table 4 of [Pástor et al. \(2022\)](#). Specifications (3), (4), (9), and (10) extend their sample period forward until December 2022. Specifications (5), (6), (11), and (12) further extend their sample backward until February 2007. Standard errors (in parentheses) are heteroskedasticity robust. *, **, and *** denote 10%, 5%, and 1% significance levels, respectively, for the null hypothesis of a zero coefficient. N refers to the total number of observations.

Figure A1: Pástor et al. (2022) pure replication



The figure shows cumulative excess returns of a green-minus-brown factor based on [Pástor et al. \(2022\)](#) and replicates their Figure 8, extending the sample period by two years until 2022. The vertically dashed line indicates the end of their sample period. The green dotted line indicates cumulative media climate change concern (“MCCC”) shocks, defined as surprises to a 36 month AR(1) process, following [Pástor et al. \(2022\)](#) using the Media Climate Change Concerns Index from [Ardia et al. \(2022\)](#). The index is available until August 2022. The MCCC time series is scaled by a factor of six for visual purposes.

Figure A2: Zhang (2025) pure replication



The figure shows cumulative excess returns of a green-minus-brown factor based on scope 1 carbon intensity as in [Zhang \(2025\)](#) and replicates her Figure 2, extending the sample period by one year from December 2021 until December 2022. The vertically dashed line indicates the end of her sample period.

level of carbon emissions (e.g., sorted on $\text{LOG}(\text{S1TOT})$). The exception is the alphas of industry-agnostic portfolios with respect to the Fama-French five-factor model augmented with momentum, but these alphas are positive, which is the opposite of what [Bolton and Kacperczyk \(2021\)](#) find. However, Tables [A2](#) and [A3](#) conduct so many tests that conventional significance tests may be misleading as we need to take multiple testing into account.

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